

Sumaiya Suravee, PhD

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Summary

Postdoctoral Researcher in Data Science and AI specialising in natural language processing (NLP), large language models, and semantic knowledge extraction. Experienced in ontology modelling, data annotation design, and building scalable AI systems, with a strong record of publications and interdisciplinary collaboration.

Work Experiences

Postdoctoral Researcher at the University of Greifswald, Germany

April 2026 - Present

- Dealing with the development and implementation of methods for automatic knowledge extraction, multimodal behaviour analysis of people with dementia.
- Designing experiments using transformer-based architecture on activity recognition datasets.
- Co-supervising doctoral students and managing the BehAlve project (Behaviour Monitoring and Support of Older Adults; see <https://datascience.uni-greifswald.de/forschung/projects/behalive/>).

Research Associate at the Greifswald University Medicine, Germany

Jan 2026 - Mar 2026

- Develops a Python package for data quality assessment for the ICD-10GM, SNOMED-CT and OPS codes.
- Works on an information retrieval task using large language models (LLM) on the histology texts, collected from the Universitätmedizin Greifswald.
- Assessed the quality of the online forum texts using a machine learning-based algorithm.

Research Associate at the University of Greifswald, Germany

Dec 2022 - Dec 2025

- Investigated automated methods for extracting knowledge from dementia-domain-specific informal forum texts.
- Developed an experimental pipeline for the named entity recognition (NER) task and relation extraction task from the dementia-based textual data using LLMs.
- Investigated the performance of the LLM for the data annotation task.
- Developed GraphRAG-based chatbot for the informal caregivers of people with dementia by incorporating domain-specific ontology.
- Developed an experimental pipeline for populating ontology and knowledge graph.

Research Associate at the University of Rostock, Germany

July 2020 - Nov 2022

- Developed eDEM-Connect: Ontology of Dementia-related Agitation and Relationship between Informal Caregivers and Persons with Dementia.
- Developed an annotation scheme and an informal dementia forum corpus where texts were scraped from dementia related textual data.

Master's Thesis Student at Planet AI GmbH, Rostock, Germany

Oct 2019 - June 2020

- Conducted literature research of several optimisers for NLP tasks such as NER systems, word embeddings, etc.
- Implemented several optimisers such as Stochastic gradient descent, Adam, Nadam, Ranger, etc., and analysed the performance of the optimiser based on the NER system by using the Conll2003 dataset.

R&D Intern at Planet AI GmbH, Rostock, Germany

April 2019 - Sep 2019

- Evaluated the performance of the NER model (Bi-directional LSTM-CNN-CRF) in Tensorflow.

- Worked for semantic word representation by using Word2Vec, Glove embedding, fastText embedding with the Conll2003 dataset.

Research Assistant at PIA care, Rostock, Germany

Oct 2018 - Mar 2019

- Worked on a project of automatic face detection for Video-based Intention recognition.
- Identified and recognised faces from given images and videos using a deep learning-based algorithm (Faster-Regional Convolution Neural Network and Single Shot Multi-box Detector).

Education

Ph.D. Candidate, University of Greifswald, Greifswald, Germany

July 2020 - March 2026

Institute of Data Science at the Department of Mathematics & Natural Science
Topic: Semantic Knowledge Extraction and Interactive Support in the Domain of Dementia

Master of Science, University of Rostock, Rostock, Germany.

Oct 2017 - June 2020

Department of Electrical Engineering (specialisation in Information Technology)

Bachelor of Science, American International University-Bangladesh, Dhaka, Bangladesh

May 2012 - Feb 2016

Department of Electrical and Electronics Engineering

Research Projects

- eDEM-CONNECT project, funded by Bundesministerium für Bildung und Forschung (BMBF) **July 2020 - Sept 2023**
- Assessing data quality aspects of texts, funded by Deutsche Forschungsgemeinschaft (DFG) **Sept 2024 - Mar 2026**
- BehAlve: Behaviour Monitoring and Support of Older Adults **April 2026 - Present**

Software Skills

Latex, Git, Tensorflow, Python, DuckDB, Large Language Models, Linux, Ontology development software (Protege software), Data annotation tools (BRAT annotation tool, Inception tool)

Publications

1. Suravee, S., Pinkert, C., Hochgraeber, I., Holle, B., Halek, M. and Yordanova, K., 2026. eDEM-CONNECT: agitation ontology for the intelligent support of informal caregivers of people with dementia. *Frontiers in aging*, 7, p.1780260.
2. Suravee, S., Razavi Ghods, M., Yordanova, K. Loc-DemCare: Supporting Informal Dementia Caregivers with a Local, Intelligent Chatbot. In *IEEE International Conference on Pervasive Computing and Communications* (Accepted)
3. Suravee, S., Schmidt, C.O. and Yordanova, K. The Challenge of Performing Ontology-driven Entity Extraction in Real-world Unstructured Textual Data from the Domain of Dementia. In *Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing - Natural Language Processing in the Generative AI Era*. pp. 1205–1214. 2025. DOI: <https://aclanthology.org/2025.ranlp-1.139>
4. Suravee, S., Yadav, D. and Yordanova, K. Relation Extraction from Real-World Unstructured Text in the Domain of Dementia. In *International Workshop on Annotation of Real-World Data for Artificial Intelligence Systems affiliated with the European Conference on Artificial Intelligence*, pp. 177–188. 2025. DOI: https://doi.org/10.1007/978-3-032-09117-8_11
5. Suravee, S., Stoev, T., Konow, S. and Yordanova, K. Assessing Large Language Models for annotating data in Dementia-Related texts: A Comparative Study with Human Annotators. In *INFORMATIK 2024, Gesellschaft für Informatik eV*. (pp. 487-498). 2024. DOI: https://doi.org/10.18420/inf2024_36

6. Suravee, S., Stoev, T., Schindler, D., Hochgraeber, I., Pinkert, C., Holle, B., Halek, M., Krüger, F. and Yordanova, K., Annotation scheme for named entity recognition and relation extraction tasks in the domain of people with dementia. In Proceedings of the International Workshop on Annotation of User Data for Pervasive Systems affiliated with the IEEE International Conference on Pervasive Computing and Communications. pp. 236-241. 2022. DOI: <https://doi.org/10.1109/PerComWorkshops53856.2022.9767278>
7. Zolfaghari, S., Suravee, S., Riboni, D. and Yordanova, K. Sensor-based locomotion data mining for supporting the diagnosis of neurodegenerative disorders: A survey. ACM Computing Surveys, 56(1), pp.1-36. 2023. DOI: <https://doi.org/10.1145/3603495>
8. Yadav, D., Suravee, S., Kemnitz, S., Strauß, T. and Yordanova, K. Prompt Engineering for Nepali NER: Leveraging Hindi-Capable LLMs for Low-Resource Languages. In Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing - Natural Language Processing in the Generative AI Era. pp. 364-1373 2025. DOI: <https://aclanthology.org/2025.ranlp-1.158>
9. Yadav, D., Suravee, S., Strauß, T. and Yordanova, K., 2024, November. Cross-Lingual Named Entity Recognition for Low-Resource Languages: A Hindi-Nepali Case Study Using Multilingual BERT Models. In Proceedings of the Fourth Workshop on Multilingual Representation Learning. pp. 167-174. 2024. DOI: <https://doi.org/10.18653/v1/2024.mrl-1.12>
10. Stoev, T., Suravee, S. and Yordanova, K. Variability of annotations over time: An experimental study in the dementia-related named entity recognition domain. In INFORMATIK 2024, Gesellschaft für Informatik eV., pp. 473-486. 2024. DOI: https://doi.org/10.18420/inf2024_35
11. Boiting, M., Tschorn, N., Suravee, S., Yordanova, K., Halek, M., Jagoda, F.A., Lüdtke, S. and Burmann, A. eDEM-CONNECT: An Ontology-Based Chatbot for Family Caregivers of People with Dementia. In International Conference on Wireless Mobile Communication and Healthcare, pp. 73-91. 2023. DOI: https://doi.org/10.1007/978-3-031-60665-6_6
12. Suravee, S., Schmidt, C.O., Yordanova, K. How Linguistic and Semantic Features Predict the Performance of Machine Learning Tasks in Informal Dementia Texts: A Named Entity Recognition Case Study. Frontiers in Digital Health (Research Topic: Healthcare Text Analytics: Unlocking the Evidence from Free Text, Volume V) (Submitted)

Professional activities

- PC member of the ARDUOUS initiative (Annotation of user Data for Ubiquitous Systems), which involves different workshops, tutorials, scientific publications and special issues.
- Volunteer at IEEE PerCom, 2022.
- Reviewer: IEEE PerCom (A* conference), ARDUOUS workshop in ECAI (A* conference).
- Supervising Master's thesis students.
- Attended Oxford machine learning summer school, University of Oxford, Oxford, England, July 2023.

Awards

- Special Award "Digitale Innovation Prozessoptimierung" at 'Inspired – Der Ideenwettbewerb in Mecklenburg Vorpommern', 2018.
- 3rd place at "Inspired – Der Ideenwettbewerb in Mecklenburg Vorpommern as Graduate and Researcher, Rostock, 2018.

Reference

- Dr. Kristina Yordanova, University of Greifswald, Greifswald, Germany, email: kristina.yordanova@uni-greifswald.de